

Variables

- **Order ID**:unique identifier for each order.
- **Distance km**:delivery distance, in kilometers.
- **Weather**:Weather conditions during delivery: Clear, Rainy, Snowy, Fog and Windy.
- **Traffic Level**:traffic conditions classified as Low, Medium or High.
- **Time_of_day**:the time the delivery took place, classified as Morning, Afternoon, Afternoon/Evening or Night.
- **Vehicle_Type**:Type of vehicle used for delivery, including Bicycle, Scooter and Car.
- **Preparation Time_min**:the time needed to prepare the order, in minutes.
- **Courier Experience yrs**:Messenger experience, in years.
- **Delivery Time_min**:Total delivery time, in minutes.

A. Remove all cases for which the variables Weather, Time_of_day and Traffic_Level have missing data. How many observations do they keep? Also report Create a table with the main descriptive statistics (minimum, mean, median, maximum, standard deviation, and number of observations) for the numerical variables. Discuss.

The following are the results obtained from the main descriptive statistics of the database after removing missing data for the variables Weather, Time_of_day and Traffic_Level:

	Mínimo	Media	Mediana	Máximo	Desvío_Estándar	N
Distance_km	0.59	10.07	10.29	19.99	5.69	912
Preparation_Time_min	5.00	16.99	17.00	29.00	7.26	912
Courier_Experience_yrs	0.00	4.64	5.00	9.00	2.92	883
Delivery_Time_min	8.00	56.45	55.00	141.00	21.58	912

After filtering the three variables, it can be observed that they all result in 912 observations, while the Courier_Experience_yrs column has 883 due to missing data (N/A). To avoid using fields with null values throughout this work and in the calculation of the different estimates, the 29 cases where the variable indicating the courier's years of experience had null data were also eliminated. Based on this, the main results for this database are presented below:

	Mínimo	Media	Mediana	Máximo	Desvío_Estándar	N
Distance_km	0.59	10.05	10.28	19.99	5.69	883
Preparation_Time_min	5.00	17.02	17.00	29.00	7.26	883
Courier_Experience_yrs	0.00	4.64	5.00	9.00	2.92	883
Delivery_Time_min	8.00	56.43	55.00	141.00	21.57	883

It can be observed that, for the distance measured in km, the mean and median are close, suggesting a fairly symmetrical distribution. On the other hand, the dispersion of 5.69 is moderate.

In the case of order preparation time, it also presents an approximately symmetrical distribution with an average dispersion of preparation times close to 7.5 minutes with respect to the mean of 17.02 minutes.

The results of the analysis of courier experience show that there are novices with zero years of experience, while there are also more experienced couriers with up to nine years. Based on the average, it can be concluded that there is a trend towards couriers having less experience, although there are cases where they deviate by almost three years from the average.

Finally, delivery time shows the greatest variability compared to the average, with a dispersion of almost 22 minutes. Although the mean and median are also close to each other, the maximum and minimum values suggest that there were deviations in some deliveries with a significant delay.

B. Often, as in this case, we work with categorical variables, so it's necessary to convert them to numeric. In machine learning, the process of converting categorical

variables into binary by creating a dummy variable for each category of the original variable is called one-hot encoding (OHE). Apply this process to each of the categorical variables in the database. How many dummy variables did you have to generate?

To resolve this section, the first step was to identify the categorical variables in the database we are working with. The selected variables were: Weather, Traffic_Level, Time_of_Day, and Vehicle_Type. For each variable, a dummy variable was created for each existing category.

- For Weather, there were 5 dummies: Clear, Foggy, Rainy, Snowy or Windy, taking a value of 1 if the category was the one indicated, 0 otherwise.
- For Traffic_Level, there were 3 dummies: High, Low or Medium, taking a value of 1 if the category was the one indicated, 0 otherwise.
- For Time_of_Day, there were 4 dummies: Afternoon, Evening, Morning or Night, taking a value of 1 if the category was the one indicated, 0 otherwise.
- For Vehicle_Type, there were 3 dummies: Bike, Car or Scooter, taking a value of 1 if the category was the one indicated, 0 otherwise.

In total, then, 15 dummies were generated to cover all the scenarios described. Within each variable, the generated dummies correspond to mutually exclusive and collectively exhaustive categories.

C. Estimate, using OLS, a model that explains delivery time as a function of distance and Weather dummies. Explain why the coefficient for one of the categories is not reported. Justify your answer. Interpret each of the estimated coefficients, commenting on the implications of statistical significance in each case.

When estimating a model including categorical variables, as is the case of Weather, the program automatically omits one of the dummy variables to avoid the problem of perfect collinearity, that is, to prevent the existence of redundant information (double accounting).

In this case, due to the way the model was presented, the category Windy was automatically omitted, so all estimated coefficients should be interpreted in comparison to the weather conditions Windy.

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Coefficients: (1 not defined because of singularities)
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  25.58523    1.60424   15.949 < 2e-16 ***
Clear        -1.47593    1.53905   -0.959  0.33783
Foggy         5.41566    1.92311    2.816  0.00497 **
Rainy         3.02055    1.69445    1.783  0.07500 .
Snowy         8.50659    1.98652    4.282 2.06e-05 ***
Windy         NA         NA         NA     NA
Distance_km   2.93270    0.07728   37.951 < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 13.01 on 877 degrees of freedom
Multiple R-squared:  0.638,    Adjusted R-squared:  0.6359
F-statistic: 309.1 on 5 and 877 DF,  p-value: < 2.2e-16

```

Given the results presented, the interpretation of the coefficients will have Windy as the "base" category and its values will be differential references with respect to the base condition.

The intercept value of 25.58 indicates that the average delivery time when the condition is Windy and the distance is 0 km is approximately 25.6 minutes (it is the value of the variable *and* in the base category) While it doesn't have a very useful interpretation in practice, it's important to mention it.

When the weather is "normal" (Clear), delivery time is reduced by 1.47 minutes compared to windy conditions (Windy). However, this variable is not statistically significant because its p-value is greater than any usual significance level. Therefore, it is possible to conclude that the difference is not significant and that the delivery time is equivalent in both weather conditions.

In the case where the weather is foggy, the delivery time increases by 5.41 minutes compared to the scenario where the weather condition is windy. This increase, unlike the previous case, is statistically significant at the 1% and 5% levels.

In the event of rainy weather, the delivery time increases by 3.02 minutes compared to the time in windy conditions, a difference that is not statistically significant with a p-value of 7.5%.

For snowy climates, there is an increase in delivery time of 8.5 minutes compared to the base category (windy) and, with a p-value very close to 0, it is a statistically significant difference.

Finally, from the analysis of the results yielded by the variable *Distance_km* it was observed that for each additional kilometer of distance, the delivery time increased by an average of 2.93 minutes. This effect was statistically significant, with a p-value almost equal to 0. This makes sense logically, as distance is expected to directly influence the delivery time of an order.

As an additional comment, and from a practical standpoint, we found it more coherent to use the "Clear" category as the base category instead of "Windy," since it is more interesting and useful to compare how delivery times vary when weather conditions are not "normal." The results for that case are shown below:

```
Residuals:
    Min       1Q   Median       3Q      Max
-33.541  -7.861  -1.182   6.638  68.723

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  24.10930    0.98225   24.545 < 2e-16 ***
Foggy         6.89158    1.45839    4.725 2.67e-06 ***
Rainy         4.49648    1.14109    3.941 8.78e-05 ***
Snowy         9.98251    1.54276    6.471 1.62e-10 ***
Windy         1.47593    1.53905    0.959  0.338
Distance_km   2.93270    0.07728   37.951 < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 13.01 on 877 degrees of freedom
Multiple R-squared:  0.638,    Adjusted R-squared:  0.6359
F-statistic: 309.1 on 5 and 877 DF,  p-value: < 2.2e-16
```

Similar to the details above, the observed coefficients should now be interpreted as the variation in delivery time for each weather condition, compared to the delivery time under

normal (clear) conditions. Since all the coefficients are positive, it is clear that delivery time increases whenever the weather worsens, whether due to fog, rain, snow, or wind.

Unlike the previous case, almost all variables are statistically significant at 0.1%.

Furthermore, the intercept decreases compared to the previous analysis, which is consistent with the fact that delivery time is shorter when the weather is clear than when it is windy. The distance in kilometers does not change since it is not dependent on weather conditions.

D. Incorporate the order preparation time into the model from the previous point. Does this improve the model's fit? Justify your answer.

The results show that including order preparation time significantly improves the model fit. The model with "Clear" omitted is included for better interpretation.

Specifically, the adjusted R^2 , which penalizes the inclusion of additional variables, increases from 0.6359 to 0.7367. This indicates that the new model explains a greater proportion of the variability in delivery time, after adjusting for the number of predictors used. Furthermore, for this new model, all variables are statistically significant at the 5% level.

```
Residuals:
    Min       1Q   Median       3Q      Max
-23.646  -6.263  -1.015   4.191  67.404

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)    7.48308    1.23244   6.072 1.88e-09 ***
Foggy          7.65677    1.24094   6.170 1.04e-09 ***
Rainy          4.72569    0.97048   4.869 1.33e-06 ***
Snowy          9.55517    1.31220   7.282 7.33e-13 ***
Windy          2.86325    1.31102   2.184  0.0292 *
Distance_km    2.96507    0.06574  45.103 < 2e-16 ***
Preparation_Time_min 0.94444    0.05147  18.348 < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 11.07 on 876 degrees of freedom
Multiple R-squared:  0.7385,    Adjusted R-squared:  0.7367
F-statistic: 412.3 on 6 and 876 DF,  p-value: < 2.2e-16
```

Likewise, the standard error of the residuals decreases from 13.01 to 11.07, which implies that the predictions of the extended model are, on average, closer to the observed values.

The coefficient associated with order preparation in the new model is positive and highly significant (coefficient = 0.9444, $p < 2e-16$), which suggests that, Keeping all other variables in the model constant, each additional minute in order preparation increases delivery time by approximately 0.94 minutes.

E. Why do you think the distance coefficient changed when we moved from the model in point (c) to (d)? Justify your answer. Support your answer with clear and precise reasoning.

When comparing model (c) with model (d), we observe a change in the coefficient associated with the variable Distance_km. This change can be explained by the incorporation of a relevant variable that had been omitted in the previous model: the order preparation time (Preparation_Time_min). The omission of this variable in model (c) probably generated

omitted variable bias, since preparation time is correlated with both distance and total delivery time.

Consequently, it is possible that the coefficient of Distance_kmin model (c) was underestimating the true impact of distance on delivery time, as it was partially absorbing the effect of the omitted variable. By including Preparation_Time_min in model (d), the effect of distance itself is better isolated, allowing for a more accurate and reliable estimate of its real impact on the dependent variable.

F. Using the model from the previous point, add traffic level dummy values and evaluate the hypothesis that, keeping all other variables constant, high traffic doubles the delay time compared to intermediate traffic. Discuss your findings.

By including these new variables, using the low traffic condition as the base category, we observed a slight improvement in the overall model fit. The adjusted R² increased to 0.7789, exceeding the values obtained in the previous models (0.7367 and 0.6359 in models d and c, respectively), which indicates a greater proportion of the variability explained by the model.

```
Residuals:
    Min       1Q   Median       3Q      Max
-20.478  -5.107  -1.219   3.327  64.365

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)    2.45888    1.22181   2.012  0.0445 *
Foggy          7.97890    1.13871   7.007 4.87e-12 ***
Rainy          4.72432    0.89019   5.307 1.41e-07 ***
Snowy          9.92914    1.20453   8.243 6.11e-16 ***
Windy          1.87860    1.20373   1.561  0.1190
Distance_km     2.96141    0.06028  49.126 < 2e-16 ***
Preparation_Time_min 0.95955    0.04718  20.336 < 2e-16 ***
High          11.87703    0.93235  12.739 < 2e-16 ***
Medium         5.95285    0.76988   7.732 2.90e-14 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 10.14 on 874 degrees of freedom
Multiple R-squared:  0.7809,    Adjusted R-squared:  0.7789
F-statistic: 389.4 on 8 and 874 DF,  p-value: < 2.2e-16
```

It is important to clarify that, in this model, with a p-value of 0.1190, the Windy category is not statistically significant. This indicates that there is no evidence to suggest that wind significantly affects delivery time. This may be due to a lack of observations under windy conditions or, among other possibilities, to low variability for this type of weather.

The reason Low was chosen as the base category was to evaluate whether, keeping the other explanatory variables constant (weather conditions, distance to the destination and preparation time), a high traffic level doubles the delay time compared to intermediate traffic.

To this end, the following hypothesis test was proposed:

$$H_0 : \beta_{High} = 2 \beta_{Medium} \quad H_1 : \beta_{High} \neq 2 \beta_{Medium}$$

This resulted in the following:

```

Linear hypothesis test:
High - 2 Medium = 0

Model 1: restricted model
Model 2: Delivery_Time_min ~ Foggy + Rainy + Snowy + Windy + Distance_km +
Preparation_Time_min + High + Medium

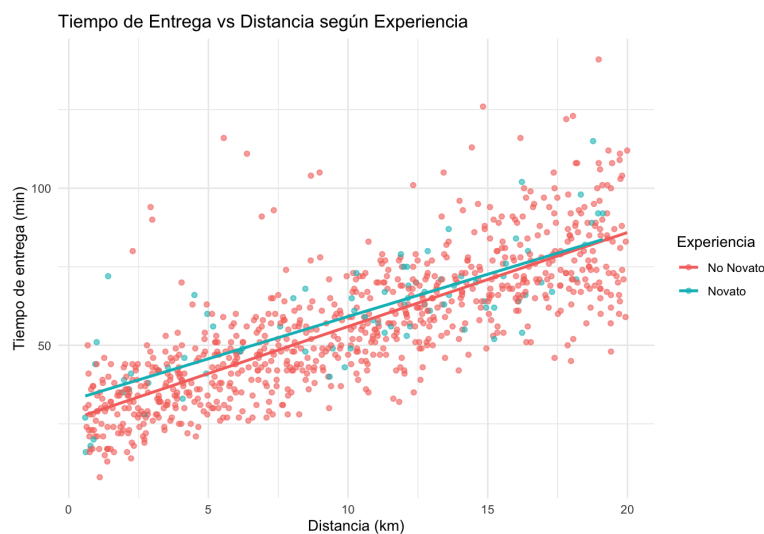
Res.Df  RSS Df Sum of Sq    F Pr(>F)
1      875 89895
2      874 89895   1  0.041477 4e-04  0.984

```

The test result yielded a p-value of 0.984, well above any significance level, which implies that there is not enough statistical evidence to reject the null hypothesis. In other words, the hypothesis that high traffic doubles the impact of average traffic on delivery time, holding all other variables constant, is not rejected.

G. Generate a dummy variable that takes the value 1 if the rider/driver has no experience, and 0 otherwise. Obtain a scatter plot of delivery time versus distance, differentiating between rider/driver experience and novice. Add regression lines for each case. Then estimate a regression that includes the generated dummy variable along with an interaction between distance and the dummy variable. Interpret the results.

After generating the dummy variable based on whether the rider was a novice or not, a scatter plot was created showing the relationship between delivery time (in minutes) and distance (in kilometers) for each condition. Regression lines were also added for each case.



For both the novice and non-novice groups, a positive relationship is observed between distance and delivery time, which is consistent with reality since, the greater the distance that must be traveled, the longer the time to do it is expected to be.

It's noticeable that the slopes of the straightaways are very similar for both groups, indicating that there wouldn't be a significant difference in how much the distance covered affects the rider's experience in terms of the time it takes to cover it. However, it's worth noting that the novice group takes slightly longer than the experienced group for shorter distances.

To complement this analysis, a regression was estimated that included the generated "Novice" dummy along with the interaction between the dummy and the distance in order to

analyze whether there are differences in the marginal effect of distance on time when the rider is a novice or not.

```

Residuals:
    Min       1Q   Median       3Q      Max
-36.143  -8.686  -1.253   7.505  73.331

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)    26.06148    0.95615   27.257  <2e-16 ***
Novato          6.20452    3.31689    1.871   0.0617 .
Distance_km     2.99236    0.08297   36.067  <2e-16 ***
Novato:Distance_km -0.30260    0.28223   -1.072   0.2839
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 13.39 on 879 degrees of freedom
Multiple R-squared:  0.6157,    Adjusted R-squared:  0.6143
F-statistic: 469.3 on 3 and 879 DF,  p-value: < 2.2e-16

```

The estimated model takes a form of the type: $Delivery_Time_min = 26.06 + 6.20 * Novato + 2.99 * Distance_km - 0.30 * (Novato * Distance_km)$.

First, from the interpretation of the intercept, it appears that the average delivery time for a non-novice rider is 26.06 minutes when the distance is 0 kilometers.

The coefficient of 6.2 associated with the Novice dummy suggests that the delivery time is 6.2 minutes longer compared to a non-novice rider, when the distance is 0 kilometers.

On the other hand, it's possible to observe that distance has a positive and significant effect on delivery time. In other words, each additional kilometer increases delivery time by approximately 2.99 minutes for experienced riders.

Finally, from the results obtained from the interaction between the novice dummy and the distance, it is concluded that it does not. This finding is statistically significant with a p-value of 0.2839. The coefficient of -0.3 indicates how the effect of distance on delivery time changes when the rider is a novice: as mentioned earlier, for non-novice riders, each additional kilometer increases delivery time by almost 3 minutes. For novice riders, that increase is $2.99 - 0.3 = 2.69$ minutes per additional kilometer.

Regarding the interaction, it is possible to conclude that the slope of the relationship between distance and time does not differ significantly depending on the rider's experience.

H. Re-estimate the model from part (d), but now express the dependent variable in logarithms. How does the interpretation of each of the coefficients change?

In this case, a logarithmic transformation was performed on the dependent variable and the interpretation of the coefficients was modified.


```

Residuals:
    Min       1Q   Median       3Q      Max
-1.05339 -0.12474  0.00493  0.11572  1.02612

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  2.9691031  0.0224921 132.006 < 2e-16 ***
Foggy        0.1646482  0.0226474   7.270 7.96e-13 ***
Rainy        0.0949941  0.0177114   5.363 1.05e-07 ***
Snowy        0.1838792  0.0239477   7.678 4.30e-14 ***
Windy        0.0569499  0.0239262   2.380  0.0175 *
Distance_km   0.0580605  0.0011998  48.393 < 2e-16 ***
Preparation_Time_min 0.0198559  0.0009394  21.137 < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.202 on 876 degrees of freedom
Multiple R-squared:  0.7684,    Adjusted R-squared:  0.7668
F-statistic: 484.4 on 6 and 876 DF,  p-value: < 2.2e-16

```

In this case, the intercept does not have a very useful interpretation since it represents the value of the $\log(\text{Delivery_Time_min})$ when all variables are 0.

Conversely, it can be observed that all coefficients take positive values, indicating that in foggy, rainy, snowy, or windy conditions, delivery time increases compared to clear weather. Furthermore, all weather effects are statistically significant.

However, this increase with respect to the base category (Clear) should be interpreted in this case as percentage increases: on foggy days, the delivery time is 16.46% longer than on clear days; on rainy days it is 9.49% longer than on clear days; on snowy days the delivery time increases by 18.38% compared to clear days and on windy days, the increase is 5.69% compared to clear days.

On the other hand, for each additional kilometer, the delivery time increases by 5.8%, while for each additional minute in order preparation, the delivery time increases by 1.9%.

In this case, due to the type of transformation performed on the dependent variable, it is not possible to compare its R^2 with that of the original model because the variances of Y and $\log(Y)$ are not comparable. As their variance was altered; their variances are not comparable. R^2 .

Anyway, looking at the R^2 Specifically for the log-linear model, it can be seen that it takes the value of 0.7684, which allows us to say that the model adequately explains the variation in the data. Furthermore, for any significance level, all variables are significant at the 5% level.

I. Incorporate a quadratic distance term into the model from point (h) above. How do you interpret the effect of distance on delivery time?

By incorporating the quadratic term of distance into the model, the marginal effect is no longer constant, but depends on it.

```

Residuals:
    Min       1Q   Median       3Q      Max
-0.94580 -0.12689  0.00629  0.10488  1.08775

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)   2.8156882   0.0270698  104.016 < 2e-16 ***
Foggy         0.1653680   0.0216164    7.650 5.29e-14 ***
Rainy         0.0983637   0.0169089    5.817 8.39e-09 ***
Snowy         0.1956095   0.0228922    8.545 < 2e-16 ***
Windy         0.0561464   0.0228371    2.459  0.0141 *
Distance_km   0.1008917   0.0047439   21.268 < 2e-16 ***
I(Distance_km^2) -0.0021146  0.0002273   -9.304 < 2e-16 ***
Preparation_Time_min 0.0200342  0.0008968   22.339 < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.1928 on 875 degrees of freedom
Multiple R-squared:  0.7893,    Adjusted R-squared:  0.7876
F-statistic: 468.1 on 7 and 875 DF,  p-value: < 2.2e-16

```

With the quadratic term included in the model, it's possible to say that, while delivery time increases with distance, this effect is reduced by the negative quadratic coefficient. That is, as distance increases, delivery time also increases, but at a decreasing rate. Therefore, the marginal effect of distance on delivery time is diminishing: each additional kilometer increases delivery time by less than the previous one.

As an additional observation, it would be possible to find the distance where the maximum delivery time is reached (in logarithms) by taking the derivative of the regression as a function of distance and setting the resulting expression equal to 0 to find the critical point.

J. Report in a single table (using the package [stargazer](#)) the results of all regressions estimated so far. Explain in which cases it is valid to compare the adjusted R² between models. Justify your answer.

Dependent variable:							
	Reg (1)	Reg1 (2)	Delivery Time Reg2 (3)	Reg3 (4)	Reg4 (5)	log(Delivery Time) Reg2.1 (6)	Reg2.2 (7)
Clear	-1.476 (1.539)						
Foggy	5.416*** (1.923)	6.892*** (1.458)	7.657*** (1.241)	7.979*** (1.139)		0.165*** (0.023)	0.165*** (0.022)
Rainy	3.021* (1.694)	4.496*** (1.141)	4.726*** (0.970)	4.724*** (0.890)		0.095*** (0.018)	0.098*** (0.017)
Snowy	8.507*** (1.987)	9.983*** (1.543)	9.555*** (1.312)	9.929*** (1.205)		0.184*** (0.024)	0.196*** (0.023)
Windy		1.476 (1.539)	2.863** (1.311)	1.879 (1.204)		0.057** (0.024)	0.056** (0.023)
Novato					6.285* (3.317)		
Distance_km	2.933*** (0.077)	2.933*** (0.077)	2.965*** (0.066)	2.961*** (0.060)	2.992*** (0.083)	0.058*** (0.001)	0.101*** (0.005)
I(Distance_km^2)							-0.002*** (0.0002)
Preparation_Time_min			0.944*** (0.051)	0.960*** (0.047)		0.020*** (0.001)	0.020*** (0.001)
High				11.877*** (0.932)			
Medium				5.953*** (0.770)			
Novato:Distance_km					-0.383 (0.282)		
Constant	25.585*** (1.604)	24.109*** (0.982)	7.483*** (1.232)	2.459** (1.222)	26.061*** (0.956)	2.969*** (0.022)	2.816*** (0.027)
Observations	883	883	883	883	883	883	883
R2	0.638	0.638	0.738	0.781	0.616	0.768	0.789
Adjusted R2	0.636	0.636	0.737	0.779	0.614	0.767	0.788
Residual Std. Error	13.015 (df = 877)	13.015 (df = 877)	11.068 (df = 876)	10.142 (df = 874)	13.394 (df = 879)	0.202 (df = 876)	0.193 (df = 875)
F Statistic	309.082*** (df = 5; 877)	309.082*** (df = 5; 877)	412.253*** (df = 6; 876)	389.396*** (df = 8; 874)	469.343*** (df = 3; 879)	484.412*** (df = 6; 876)	468.132*** (df = 7; 875)
Note: *p<0.1; **p<0.05; ***p<0.01							

In this case, comparing the result it yields is valid. R^2 adjusted between models only when the models have the same dependent variable and the regressors vary: the comparison of the R^2 . The adjustment is made to penalize for the variables that are added to the model.

In that sense, for the reg to reg4 models, it is correct and useful to compare their R^2 adjusted since they have the same dependent variable (Delivery_Time_min).

For log-lin models (reg2.1 and reg2.2), however, the following should not be compared R^2 with the previous models since the dependent variable was modified logarithmically and, therefore, its scale differs from the others.

He R^2 It is a measure that indicates the proportion of the variable and (Delivery time) is being explained by the model. Thus, the R^2 tight ($\bar{R}^2$) allows correct for degrees of freedom and compare the fit between two or more models that have different numbers of predictors on the same variable and. Therefore, in its calculation, it includes a penalty factor that depends on the sample size and the number of predictors in the model..

In conclusion, it is a metric that allows comparing the predictive power of two or more models and the effect will depend on the power of the additional regressor. If I add a regressor and $\bar{R}^2$ If it doesn't change or decrease, the regressor I added didn't make a significant contribution. On the contrary, the relative contribution of the additional variable is useful in explaining the variability of the delivery time.

Based on this logic, in the following section, the best and worst fitting models will be selected.

K. Choose the best-fit and worst-fit models from the non-logarithmic models. Using both models, predict the delivery time for all observations in the sample and calculate the prediction errors. Report the main descriptive statistics for the prediction errors of each model, and calculate the RMSE and MAPE. What do you observe?

When comparing models reg3 and reg4, it is observed that the former presents a better overall fit than the latter, which is reflected in the comparison of their adjusted coefficients of determination and the quantity and relevance of the regressors included.

In particular, model reg3 achieves an adjusted R^2 of 0.779, indicating a high explanatory power of the model: approximately 78% of the observed variability in delivery time is explained by the incorporated variables. In contrast, reg4 shows a considerably lower adjusted R^2 of 0.614, making it the worst of all, demonstrating a smaller proportion of variability explained by the model and, therefore, a much more limited fit.

To evaluate the predictive quality of the models, prediction errors were analyzed, defined as the difference between the actual delivery time and the time estimated by each model. The main descriptive statistics for these errors are presented below.

```

> summary(error_mejor)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
-20.478 -5.107  -1.219   0.000   3.327  64.365
> summary(error_peor)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
-36.143 -8.686  -1.253   0.000   7.505  73.331

```

For reg3, it is observed that the mean of the errors is 0, indicating that, on average, the model does not tend to underestimate or overestimate delivery times.

Given that the median error value is -1.219, there is a slight tendency to overestimate delivery times. Furthermore, the interquartile range extends from -5.107 to 3.327 minutes, indicating moderate variability. The extreme error values range from -20.478 to 64.365 minutes, without showing excessive outliers or serious atypical behavior.

In contrast, reg4 also has a mean of 0, indicating that the model is not biased on average. With a median of -1.253, very similar to that of reg3, it also indicates a slight average overestimation.

However, the dispersion is considerably greater: the first quartile is at -8.686 and the third at 7.505, giving a higher interquartile range than that of reg3. This represents greater variability in the predictions.

Furthermore, the extreme values are wider, with a minimum of -36,143 and a maximum of 73,331 minutes, which also reflects greater instability in the predictions..

On the other hand, reg3 and reg4 were evaluated based on the Mean Squared Error (RMSE) and the Mean Absolute Percentage Error (MAPE).

rmse_mejor	10.0899269546447	mape_mejor	10.9848811177074
rmse_peor	13.3638675418492	mape_peor	20.2333035566808

Regarding the model in reg3, previously identified as the most accurate in terms of error dispersion, it also exhibits better fit metrics. Its RMSE is 10.09 minutes, significantly lower than the RMSE of 13.36 minutes observed for reg4. This difference implies that, on average, reg3's predictions deviate less from the true value, with larger errors being particularly penalized due to the quadratic nature of the metric.

Furthermore, the MAPE for reg3 is 10.98%, compared to 20.23% for reg4. This result indicates that, on average, the prediction errors of the former represent approximately 11% of the actual value, while in the latter, the errors reach over 20%. Since MAPE expresses the magnitude of the relative error, reg3 provides closer and more consistent estimates in proportional terms, regardless of the actual lead time.

L. Reshuffling. Starting with the best-fit model from the previous point, we want to conduct a placebo test to demonstrate that the effect of distance on delivery time does not exist when it shouldn't. How do we do this? The goal is to perform an experiment based on the observed data, in which the values of the distance variable (only) between orders are randomly permuted many times (i.e., 10,000 times), estimate

the regression for each permutation, and compare the coefficients and p-values of the distance variable with those observed in the regression with the original sample. Implement this test, comment on the results, and explain the intuition behind this approach..

To validate the robustness of the observed effect of distance on delivery time, a permutation test was implemented. This involves randomly permuting the values of the variable `Distance_km` between observations, so that any possible real association between said variable and the dependent variable is broken (`Delivery_Time_min`). From this rearrangement, the same regression model used previously was estimated, repeating this procedure 10,000 times. In each iteration, the coefficient associated with `Distance_km`.

```
> # Modelo de mejor ajuste:
> reg3 <- lm(Delivery_Time_min ~ Foggy + Rainy + Snowy + Windy + Distance_km +
+           Preparation_Time_min + High + Medium, data = datos_filtrados)
>
> # Se guarda el coeficiente de la distancia del modelo de mejor ajuste
> coef_mejor <- coef(reg3)["Distance_km"]
>
> coef_simulados <- c()
>
> # Se repite el proceso 10000 veces
> set.seed(123)
> for (i in 1:10000) {
+   datos_simul <- datos_filtrados
+   datos_simul$Distance_km <- sample(datos_filtrados$Distance_km)
+   modelo_simulado <- lm(Delivery_Time_min ~ Foggy + Rainy + Snowy + Windy + Distance_km +
+                         Preparation_Time_min + High + Medium, data = datos_simul)
+   coef_simulados[i] <- coef(modelo_simulado)["Distance_km"]
+ }
```

Next, an empirical p-value was calculated as the proportion of simulations in which the absolute value of the permuted coefficient was greater than or equal to the absolute value of the coefficient estimated in the original model.

```
> # Se calcula el p-valor empírico: cuántos coeficientes simulados son tan extremos como el real
>
> p_empirico <- mean(abs(coef_simulados) >= abs(coef_mejor))
> print(p_empirico)
[1] 0
```

The result was an empirical p-value of 0, indicating that none of the 10,000 permutations produced a coefficient as extreme as that observed in the actual data. This result provides strong evidence that the estimated effect of distance on delivery time is not attributable to chance. In other words, the relationship observed in the original model reflects a systematic pattern in the data, and not a result influenced by random correlation.